RESEARCH ARTICLE

Temporal Trends in the Frequency of Concomitant Cancer and Cerebrovascular Disease Diagnoses in the United States

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Abstract

Background and Objectives

Cancer is associated with vascular disease. Modern cancer treatments, particularly targeted agents and immunotherapy, have extended survival from cancer. We aimed to determine whether the co-prevalence of cancer and cerebrovascular disease diagnoses among outpatients has increased over time along with these advances.

Methods

This was a time-series cross-sectional analysis of outpatient visits in the US National Ambulatory Medical Care Survey from 2005 to 2019. The primary cohort included patient visits with a diagnosis of cerebrovascular disease (ischemic or hemorrhagic stroke or TIA). We used joinpoint regression to evaluate temporal trends in the frequency of concomitant cancer diagnoses during the study period. We used logistic regression, adjusting for demographics and stroke risk factors, to assess the relationship between survey year and cancer diagnoses.

Results

After applying sampling weights, this analysis represented an estimated 245,093,640 visits by patients with a cerebrovascular disease diagnosis. The mean patient age for the overall cohort was 69.4 years (SD, 15.9), and 51.0% were women. Among patients with cerebrovascular disease, 27,294,617 visits (11.1%) included a concomitant diagnosis of cancer. The percentage of visits listing a concomitant cancer diagnosis increased from 7.7% (95% CI 4.6%–10.7%) in 2005 to 15.3% (95% CI, 6.1%–24.5%) in 2019. Joinpoint regression analysis demonstrated a significant increase in cancer diagnoses from 2005 to 2019, with an annual percentage change of 3.5% (95% CI, 0.2%–6.9%). In multivariable logistic regression analysis, the co-prevalence of cancer and cerebrovascular disease increased linearly over time (adjusted odds ratio per year, 1.04; 95% CI, 1.00–1.08).

Discussion

From 2005 to 2019, the proportion of US ambulatory visits with concomitant diagnoses of cerebrovascular disease and cancer steadily increased from 7.7% to 15.3%. Because data on the timing of diagnoses are unavailable in NAMCS, this analysis could not determine whether these diagnoses were related, and which came first. Future population-based studies including data on the timing and etiology of stroke and cancer diagnoses are needed to better elucidate the potential causal link between these diseases.

Introduction

In the United States, the prevalence of stroke has increased since 2011.[1] This is at least in part related to temporal reductions in stroke mortality resulting from advances in stroke systems of

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Table 1 Characteristics of Patient Visits With Diagnoses of Cerebrovascular Disease in NAMCS, 2005–2019, Stratified by Diagnoses of Cancer

Characteristic[a,b]	Diagnosis of cancer (n = 27,294,617)	No diagnosis of cancer (n = 217,799,023)	*p* Value
Demographics			
Age, y	74.2 (12.8)	68.7 (16.0)	<0.001
Sex, (%)			0.515
Female	49.1	51.3	
Male	50.9	48.7	
Race (%)			0.786
Black	9.1	10.2	
White	84.4	84.7	
Other	6.5	5.1	
Metropolitan residence	11.1	11.6	0.784
Medical comorbidities (%)			
Chronic kidney disease[c]	13.2	10.8	0.608
COPD	13.7	10.0	0.045
Coronary artery disease[c]	28.0	26.4	0.816
Diabetes mellitus	26.2	28.2	0.435
Hyperlipidemia	40.3	45.9	0.072
Hypertension	72.7	67.4	0.058
Obesity	11.9	10.7	0.662
Obstructive sleep apnea[c]	7.6	10.4	0.442
Venous thromboembolism[c]	7.6	2.6	0.084
Body mass index	28.6 (7.5)	28.5 (6.3)	0.848

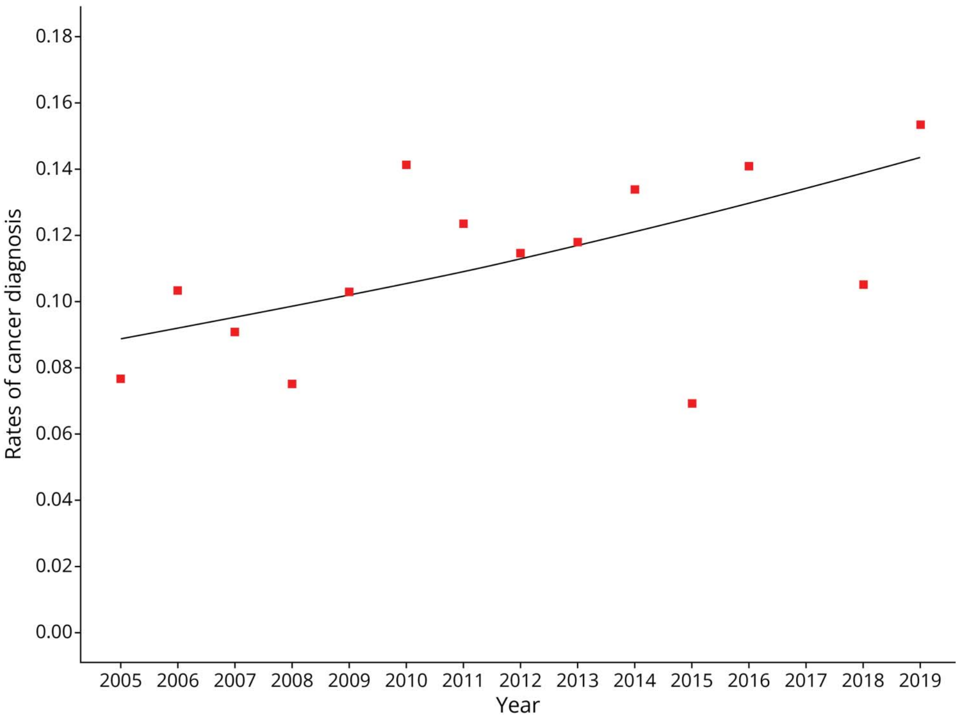
Abbreviations: COPD = chronic obstructive pulmonary disease; NAMCS = National Ambulatory Medical Care Survey.
[a] Data reflect the estimated number of representative visits after applying sampling weights.
[b] Data reported as % for categorical variables and mean (SD) for continuous and ordinal variables. Percentages are rounded and may add up to >100%.
[c] Data only available for years 2014 through 2019.

care, quality processes, and acute and secondary prevention treatments.[2,3] In addition, because cancer and its treatments are risk factors of stroke, longer patient survival from cancer could also be contributing to this increased stroke prevalence.[4,5] Increased screening and improved treatments have led to earlier diagnoses and extended survival from cancer, with more than two-thirds of US patients now surviving more than 5 years after initial cancer diagnosis.[6,7] Immunotherapy and targeted agents, such as small-molecule pathway inhibitors and antibody-drug conjugates, have played important roles in this advance.[8]

Patients with cancer face a substantially increased risk of stroke through cancer-mediated hypercoagulability, shared risk factors such as smoking and obesity, adverse effects of cancer treatments, and frequent interruptions in antithrombotic medications.[4,5] Pathways implicated in the multifactorial prothrombotic state associated with cancer include secreted procoagulant factors such as tissue factor, platelet activation, endothelial dysfunction, neutrophil extracellular trap formation, and circulating cancer and platelet microvesicles.[9] Some cancer drugs may also increase stroke risk through endothelial injury or other mechanisms, including vascular endothelial growth factor agents, tyrosine kinase inhibitors, platinum-based compounds, and hormonal agents.[9] In addition, radiotherapy heightens long-term stroke risk by promoting a progressive large and small vessel vasculopathy within treated areas.[5,10,11]

Given these evolving epidemiologic and pathophysiologic considerations, we used the National Ambulatory Medical Care Survey (NAMCS) to estimate the frequency and temporal trends of concomitant cancer and stroke diagnoses among ambulatory patient visits in the United States from

Figure 1 Rates of Cancer Diagnosis Among Patient Visits With a History of Cerebrovascular Disease in NAMCS, 2005–2019



Joinpoint regression graph demonstrating rates of cancer diagnosis by survey year among patient visits with diagnoses of cerebrovascular disease in NAMCS from 2005 through 2019 (estimated n = 245,093,640 after applying sampling weights). The annual percentage change was 3.5% (95% CI 0.2%–6.9%; p = 0.038). Annual rates are displayed as red boxes. NAMCS = National Ambulatory Medical Care Survey.

2005 to 2019. We hypothesized that, during this time, the frequency of cancer diagnoses steadily increased among patients with concomitant cerebrovascular disease.

Methods

Design

We performed a time-series cross-sectional analysis of patients enrolled in NAMCS between 2005 and 2019 (omitting 2017 because no survey data have been publicly released for that year because of delays by the National Center for Health Statistics in harmonizing and integrating collected data from different sources). NAMCS is a nationwide, annual, probability sample survey that uses prospective and retrospective data collection methods to report the characteristics of patient visits using ambulatory medical care services in the United States.[12] NAMCS projects ambulatory care utilization data based on a sample of patient visits to community health centers and non–federal government physicians engaged in office-based patient care.[12] We excluded years after 2019 because NAMCS stopped collecting office-based visit data during the height of the COVID-19 pandemic. In addition, although NAMCS has released some visit data from years 2020 to 2023, these data would not be generalizable because of the COVID-19 pandemic and its documented effects on stroke risk and health care utilization.[13,14]

Population

Since 2005, participating physicians have systematically reported whether survey participants have diagnoses of cancer or cerebrovascular disease. The population for this analysis included participants of any age with cerebrovascular disease, ascertained in NAMCS as follows: "Does patient now have cerebrovascular disease/history or stroke, or TIA?" Therefore, for this analysis, cerebrovascular disease is broadly defined as any history of ischemic stroke, hemorrhagic stroke, TIA, or other forms of cerebrovascular disease, regardless of mode of diagnosis, severity, mechanism, or timing.

Measurements

The study outcome was a diagnosis of cancer, ascertained in NAMCS as follows: "Does patient now have cancer?" Study covariates included participants' age, sex, race, residential status (metropolitan vs nonmetropolitan), body mass index, and history of the following stroke risk factors: chronic kidney disease, chronic obstructive pulmonary disease (COPD), coronary artery disease, diabetes mellitus, hyperlipidemia, hypertension, obesity, venous thromboembolism, or obstructive sleep apnea. Data on atrial fibrillation are unavailable in NAMCS.

Analyses

Descriptive statistics with sampling weights for survey data were used to estimate participants' baseline characteristics.

Table 2 Estimated Number and Percentage of Patient Visits With Diagnoses of Cerebrovascular Disease and Cancer in NAMCS by Calendar Year

Year[a]	Number of visits with cerebrovascular disease diagnoses	Number of visits with cancer diagnoses	Among visits with cerebrovascular disease diagnoses, rate (95% CI) of concomitant cancer diagnoses
2005	18,318,683	1,403,866	7.7% (4.6%–10.7%)
2006	14,543,673	1,502,413	10.3% (6.5%–14.2%)
2007	15,286,322	1,387,844	9.1% (6.5%–11.7%)
2008	16,364,812	1,228,191	7.5% (4.9%–10.1%)
2009	19,481,413	2,003,582	10.3% (6.4%–14.1%)
2010	13,348,930	1,885,960	14.1% (9.4%–18.9%)
2011	20,083,079	2,480,830	12.4% (8.1%–16.6%)
2012	17,505,641	2,004,944	11.5% (9.3%–13.6%)
2013	18,799,363	2,216,383	11.8% (8.3%–15.3%)
2014	15,663,430	2,096,499	13.4% (9.9%–16.8%)
2015	17,403,515	1,205,001	6.9% (3.9%–10.0%)
2016	16,816,766	2,368,835	14.1% (8.9%–19.2%)
2017[b]	—	—	—
2018	17,638,842	1,853,109	10.5% (4.1%–16.9%)
2019	23,839,172	3,657,160	15.3% (6.1%–24.5%)

Abbreviation: NAMCS = National Ambulatory Medical Care Survey.
[a] Data reflect the estimated number of representative visits after applying sampling weights.
[b] NAMCS data from 2017 have not been publicly released.

The sample weight (patient visit weight) reflected the number of visits in the US population represented by a sampled visit. Visit weights included inflation by reciprocals of selection probabilities, adjustment for nonresponse and population ratios, and weight smoothing to mitigate potential biases common to survey data. Following NAMCS instructions, we applied the sample weight to produce national estimates and accurately assess sampling error for all statistics reported in this study. The Student *t* test and the χ^2 test were used to compare continuous and categorical variables between participant visits with and without a diagnosis of cancer. We used joinpoint regression to evaluate whether the frequency of cancer has increased over time in patients with cerebrovascular disease diagnoses, to estimate the annual percent change during the study period, and to determine whether there was a change point (i.e., a specific point in time when the trend of a time-series analysis significantly changes in direction or slope). To further investigate the relationship between time and the frequency of cancer diagnoses, we performed univariable and multivariable logistic regression with survey year as the exposure of interest. The multivariable models adjusted for age, sex, race, metropolitan status, and stroke risk factors. In secondary analysis, we evaluated the co-frequency of cancer and cerebrovascular disease diagnoses among all participant visits from 2005 to 2019. Statistical significance was set at $p < 0.05$.

Standard Protocol Approvals, Registrations, and Participant Consents

Weill Cornell's Institutional Review Board certified this analysis of deidentified data as exempt from review and waived participant consent requirements.

Data Availability

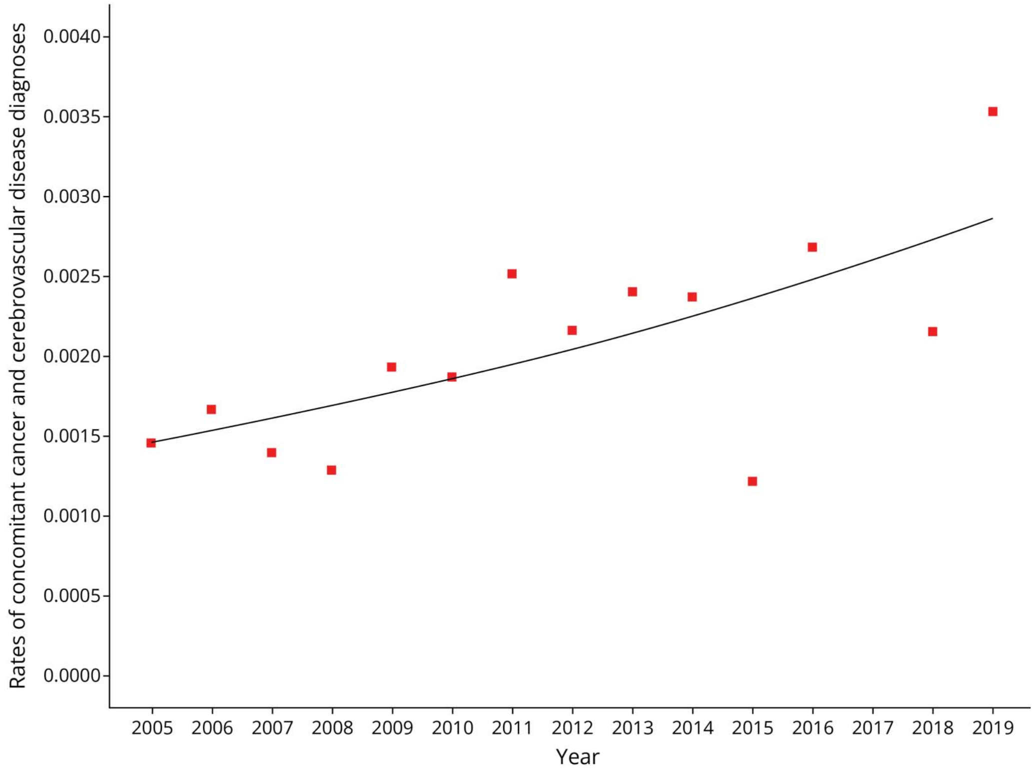
NAMCS data are publicly available online.[15]

Results

Characteristics

After applying sampling weights, this analysis represented an estimated 245,093,640 patient visits with a recorded diagnosis of cerebrovascular disease. The mean patient age for the overall cohort was 69.4 years (SD, 15.9), and 51.0% were women. Among patients with cerebrovascular disease, 27,294,617 visits (11.1%) included a concomitant diagnosis of cancer. Compared with patient visits without a diagnosis of cancer, patient visits with a diagnosis of cancer included

Figure 2 Rates of Concomitant Cancer and Cerebrovascular Disease Diagnoses Among All Patient Visits Surveyed in NAMCS, 2005–2019



Joinpoint regression graph demonstrating rates of concomitant cancer and cerebrovascular diseases diagnoses by calendar year among all patient visits surveyed in NAMCS from 2005 through 2019 (estimated n = 13,356,826,182 after applying sampling weights). The annual percentage change was 4.9% (95% CI 0.6%–9.7%; $p = 0.025$). Annual rates are displayed as red boxes. NAMCS = National Ambulatory Medical Care Survey.

significantly older patients and higher rates of COPD (Table 1).

Primary Analyses

Among patient visits with a diagnosis of cerebrovascular disease, the percentage of visits with a concomitant diagnosis of cancer increased from 7.7% (95% CI, 4.6%–10.7%) in 2005 to 15.3% (95% CI, 6.1%–24.5%) in 2019 (Figure 1, Table 2). In joinpoint regression analysis, the percentage of visits with a diagnosis of cancer increased over time, with a significant annual percentage change of 3.5% (95% CI, 0.2%–6.9%; $p = 0.038$). A change point was not detected.

In univariable logistic regression analysis, the co-prevalence of cancer and cerebrovascular disease diagnoses increased linearly, albeit nonsignificantly, over time (OR/year, 1.03; 95% CI, 0.99–1.08; $p = 0.093$). After controlling for age, sex, race, metropolitan status, and stroke risk factors, the adjusted OR between survey year and a diagnosis of cancer was significant at 1.04 (95% CI, 1.00–1.08; $p = 0.048$).

Secondary Analyses

When evaluating all participant visits in NAMCS from 2005 to 2019 (estimated n = 13,356,826,182), the percentage of visits with diagnoses of both cancer and cerebrovascular disease increased from 0.15% (95% CI, 0.07%–0.22%) in 2005 to 0.35% (95% CI, 0.12%–0.58%) in 2019 (Figure 2). In joinpoint regression analysis, the percentage of visits with diagnoses of both cancer and cerebrovascular disease increased over time, with a significant annual percentage change of 4.9% (95% CI, 0.6%–9.7%; $p = 0.025$). A change point was not detected.

Discussion

Using ambulatory survey data representing over 245 million US patient visits, we found that among patients with stroke/TIA diagnoses, the frequency of a concomitant cancer diagnosis steadily increased from 7.7% in 2005 to 15.3% in 2019, with an annual percentage change of 3.5% and no clear inflection point. This increased co-frequency persisted after controlling for demographics and stroke risk factors. When evaluating all patient visits in NAMCS, the frequency of visits with diagnoses of both cerebrovascular disease and cancer increased from 0.15% in 2005 to 0.35% in 2019, indicating that, in modern times, approximately 1 in 285 outpatient visits in the United States involves both conditions. The

TAKE-HOME POINTS

- → In an analysis of the National Ambulatory Medical Care Survey representing an estimated 245,093,640 patient visits with a diagnosis of cerebrovascular disease, 27,294,617 visits (11.1%) included a concomitant diagnosis of cancer.
- → The frequency of a concomitant cancer diagnosis increased from 7.7% in 2005 to 15.3% in 2019, with an annual percentage change of 3.5% and no clear inflection point.
- → Clinicians should be aware that the co-frequency of cancer and cerebrovascular disease in the US outpatient setting has greatly increased over time.

estimated annual percent increase in visits with both diagnoses was numerically but not significantly higher when evaluating all NAMCS patient visits (4.9%) instead of just visits with recorded cerebrovascular disease diagnoses (3.5%). Although this may represent chance, as supported by the overlapping CIs, this also could reflect temporal declines in mortality rates for both diseases.[3,6,7]

To the authors' knowledge, no previous study has evaluated the frequency of cancer among outpatients with cerebrovascular disease. However, previous studies have evaluated the prevalence of cancer diagnoses among hospitalized patients with cerebrovascular disease, and those studies' findings are consistent with ours.[16,17] In a National Inpatient Sample analysis representing >5 million patients hospitalized in the United States with a primary discharge diagnosis of acute ischemic stroke, the prevalence of a concomitant cancer diagnosis increased from 10.6% in 2007 to 14.0% in 2019 (annual percentage change of 1.7%).[16] In a prospective study of 2,736 patients diagnosed with acute ischemic stroke or TIA at 8 Dutch hospitals from 2007 to 2014, 12% had a cancer history.[17] Our study differs from these analyses by focusing on ambulatory patients who have survived their cerebrovascular event, thereby demonstrating that despite the high mortality rates in patients with cancer-associated stroke,[18,19] the co-frequency of cancer and cerebrovascular disease in the community has substantially increased over time. These data highlight the growing clinical role that cancer and its treatments now play in the diagnosis and management of patients with stroke.[5]

This study has limitations. First, it lacked information on important clinical characteristics, including the type, extent, and activity of cancers and the type and mechanism of cerebrovascular events, because these data were not reliably collected in NAMCS. This prevented us from differentiating ischemic stroke vs hemorrhagic stroke vs TIA, which could have led to diagnostic bias, especially as TIA is commonly misdiagnosed.[20] Furthermore, the broad composite definition of cerebrovascular disease in NAMCS could have diluted the cohort of true stroke or TIA survivors by including cerebrovascular-related conditions such as aneurysms or carotid disease. In addition, data on atrial fibrillation, a disease linked to both cancer and stroke, were unavailable.[21-23] Nevertheless, it is also important to highlight the strengths of the NAMCS data set and why we used it for this analysis, namely that it is prospectively collected, nationally representative, and large in size and statistical power. Second, owing to its cross-sectional design and the lack of data on the timing of diagnoses, we were unable to evaluate whether the cancer and cerebrovascular diagnoses were related, and which came first. It is possible that the observed increased frequency of cancer diagnoses in our study could be related to temporal improvements in cancer screening programs and diagnostic technologies or be reflective of potential changes in health care utilization or outpatient access. Third, since its inception, NAMCS has undergone several changes in its survey questions and data collection methods, and these changes could have affected the rates of recorded cancer diagnoses. Similarly, given temporal improvements in survivorship from both cancer and stroke,[3,6] providers' documentation and diagnosis recording practices could have changed over time, and this could have confounded observed trends. Fourth, NAMCS is restricted to ambulatory patient visits to nonfederally employed physicians in the United States. Therefore, patients who had severe strokes that led to early death were excluded, which likely skewed our population toward patients with milder cerebrovascular events. Fifth, the annual number of patient visits in NAMCS varies year to year according to survey response rates and primary care utilization in the United States, and these patterns could have affected cancer diagnosis rates.[12] Physician response rates in particular are influenced by a variety of factors, including their specialty, geographic location, and practice setting. NAMCS adjusts for nonresponse rates in their visit weights; however, nonresponse bias is still possible and could have affected our results.

From 2005 to 2019, the frequency of cancer diagnoses among ambulatory patients with a history of stroke or TIA in the United States steadily increased from 7.7% to 15.3%. These findings should be viewed as hypothesis generating because of several limitations from our cross-sectional survey-based study, including the lack of granular data on the type and timing of cancer and cerebrovascular disease diagnoses and the potential for diagnostic and nonresponse biases and measurement error. Future longitudinal studies with specific cancer and stroke diagnostic data from individual patients are needed to confirm these findings. In the meantime, clinicians treating patients with cancer should be aware of the potential temporal increase in concomitant cerebrovascular diagnoses, and there should be renewed emphasis on the prevention of stroke in this population. Clinical trials are needed to determine the optimal strategies to prevent cerebrovascular events in patients with cancer.

Author Contributions

M. Beyeler: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data. L. Lanphier: drafting/revision of the manuscript for content, including medical writing for content. C. Zhang: study concept or design; analysis or interpretation of data. A.L. Liberman: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. S.T. Tagawa: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. S. Mantha: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. H. Kamel: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. B.B. Navi: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data.

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